SSEE-V3.6 and the Hubble Tension: An Algebraic Screening Fraction from First Principles

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Abstract

The Hubble tension between the CMB-inferred value $H_0^{\rm CMB} = 67.4 \pm 0.5$ km/s/Mpc and the local distance-ladder measurement $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ km/s/Mpc remains one of the sharpest anomalies in modern cosmology. Within the Structural Self-Energy Expansion (SSEE-V3.6) framework, the algebraic constants φ (golden ratio) and π determine all parameters with zero free parameters. We derive algebraically that the dark-energy EFT of Paper 7 and the disformal geodesic of Paper 8 together imply a *local screening fraction*

$$f_{\rm screen} = \frac{\alpha_K}{3\,\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} \approx 0.0673\,,$$

where $\alpha_K = 3\,\Omega_{\rm DE}(1 + w_0)$ is the EFT kineticity derived from the background equation of state, and $\mathcal{M} = \beta_c/2 = (3\varphi + \pi)/4 \approx 1.9989$ is the effective optical coupling derived from the disformal null geodesic. The two structural constants $\mathcal{A} = 2\mathcal{M}$ cancel exactly in the ratio, leaving a pure function of φ and π. Applied as a local correction to the cosmological value $H_0^{\rm alg} = 3(\varphi + \pi)^2$,

$$H_0^{\rm local} = \frac{H_0^{\rm alg}}{1 - f_{\rm screen}} = \frac{67.96}{1 - 0.0673} \approx 72.86 \text{ km/s/Mpc}\,,$$

which lies 0.17σ from SH0ES (Riess et al. 2022) with *zero free parameters*. The correction operates at late times ($z < 0.15$) through the scalar kinetic energy contribution to the local expansion rate, and does not modify the sound horizon r_d, the CMB peak positions, or the growth factor σ_8 derived in Papers 3 and 5.

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1 Introduction

The Hubble tension [Verde et al., 2019, Riess et al., 2022, Kamionkowski & Riess, 2023] is the 4–6σ discrepancy between measurements of the present expansion rate obtained by two independent methods:

- **Early-universe inference**: CMB anisotropies (Planck PR4, Planck Collaboration 2020) combined with acoustic oscillation physics yield $H_0 = 67.4 \pm 0.5$ km/s/Mpc under the assumption of ΛCDM.
- **Local distance-ladder**: Type Ia supernovae calibrated with Cepheids in the SH0ES programme [Riess et al., 2022] give $H_0 = 73.04 \pm 1.04$ km/s/Mpc, probing predominantly $z < 0.15$.

The SSEE-V3.6 framework [Almeida, Paper1,P,P,P] derives the cosmological expansion rate algebraically:

$$H_0^{\mathrm{alg}} \;=\; 3(\varphi + \pi)^2 \;\approx\; 67.96 \text{ km/s/Mpc}\,, \tag{1}$$

which agrees with Planck at the 0.81σ level (MCMC Paper 2, Almeida Paper2) but does not, at first sight, resolve the local tension.

In this paper we show that the same algebraic structure that produces H_0^{alg} also produces a local kinematic correction $f_{\mathrm{screen}} \approx 0.0673$ from first principles, predicting $H_0^{\mathrm{local}} = 72.86$ km/s/Mpc. The derivation combines two results already established in this series:

1. From Paper 7 [Almeida, Paper7]: the EFT kineticity $\alpha_K = 3\,\Omega_{\mathrm{DE}}(1 + w_0)$, which follows from the background Friedmann equations and the k-essence action $K(X) = X/\mathrm{KAL}_0$.
2. From Paper 8 [Almeida, Paper8]: the effective optical coupling $\mathcal{M} = \beta_c/2 = (3\varphi + \pi)/4$, which emerges from the disformal null geodesic of dark matter.

Together these give the exact algebraic identity $f_{\mathrm{screen}} = \alpha_K/(3\mathcal{M})$.

Organisation. Section 2 recalls the key results from Papers 7 and 8. Section 3 derives the algebraic identity in three steps. Section 4 interprets the result physically. Section 5 verifies consistency with the rest of the SSEE series. Section 6 states the pre-committed falsifiable prediction and its current observational status. Section 8 summarises.

2 Background: EFT Kineticity and Disformal Coupling

2.1 The SSEE action (Paper 7)

The fundamental action of SSEE-V3.6 is a k-essence scalar minimally coupled to gravity and conformally coupled to dark matter:

$$S = \int d^4x\,\sqrt{-g}\,\left[\frac{M_{\mathrm{Pl}}^2}{2}R + K(X,\phi) + \mathcal{L}_m\right]\,, \tag{2}$$

with kinetic kernel

$$K(X,\phi) = \frac{X}{\mathrm{KAL}_0} + \frac{X^2}{M^4} - V_0\, e^{-\lambda\phi/M_{\mathrm{Pl}}}\,, \tag{3}$$

where $X = -\frac{1}{2} g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$ and

$$\mathrm{KAL}_0 = \frac{\varphi+\pi}{2} + \pi \approx 5.521\,, \tag{4}$$

$$\beta_c \equiv \mathcal{A} = \frac{3\varphi+\pi}{2} \approx 3.998\,. \tag{5}$$

All parameters are algebraically fixed; no fitting is performed.

The dark-energy equation-of-state parameters derived in Paper 1 are

$$w_0 = -\frac{\mathcal{A}}{\varphi+\pi} \approx -0.840\,, \tag{6}$$

$$\Omega_{\mathrm{DE}} = -w_0 = \frac{\mathcal{A}}{\varphi+\pi} \approx 0.840\,. \tag{7}$$

The EFT kineticity in the Bellini–Sawicki classification [Bellini & Sawicki, 2015] is defined as

$$\alpha_K \;\equiv\; \frac{2X\,K_X}{M_{\mathrm{Pl}}^2 H^2}\,, \tag{8}$$

where $K_X = \partial K/\partial X$. Paper 7 derives $\alpha_K = 0.4033$ numerically from CLASS/EFTCAMB; in Section 3 we provide its exact algebraic expression.

2.2 The disformal coupling and MIRA (Paper 8)

Paper 8 extends the action to the strong-gravity regime by coupling dark matter to the disformal metric

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} + \frac{2}{M^4}\,\partial_\mu\phi\,\partial_\nu\phi\,. \tag{9}$$

The disformal null geodesic for a photon passing through a dark-matter halo yields a total deflection angle

$$\hat{\alpha}_{\mathrm{total}} \approx \beta_c \left(\frac{4GM}{c^2 b}\right)\,, \tag{10}$$

where b is the impact parameter and $\beta_c = \mathcal{A}$. The effective Einstein ring radius is therefore amplified by

$$\frac{\theta_E^{\mathrm{SSEE}}}{\theta_E^{\mathrm{GR}}} \approx \sqrt{\beta_c} \approx \mathcal{M}\,, \qquad \mathcal{M} \equiv \frac{\beta_c}{2} = \frac{3\varphi+\pi}{4} \approx 1.9989\,. \tag{11}$$

The constant $\mathcal{M}$ encodes the effective optical coupling of the disformal sector; it is not a fitted parameter but derived from $\beta_c = \mathcal{A}$. Paper 8 further derives $\mathcal{M} = \Omega_{\mathrm{DE}}^{\mathrm{CMB}}/\Omega_{\mathrm{DE}}^{\mathrm{dyn}}$, confirming its role as a structural ratio between the CMB-era and dynamical matter fractions.

3 Algebraic Derivation: $f_{\mathrm{screen}} = \alpha_K/(3\mathcal{M})$

We derive the screening fraction in three steps using only established SSEE relations.

3.1 Step 1: Exact expression for α_K

The background Friedmann equations give

$$\rho_\phi = \frac{X}{\mathrm{KAL}_0} + V = 3M_{\rm Pl}^2 H^2\, \Omega_{\rm DE}\,, \tag{12}$$

$$p_\phi = \frac{X}{\mathrm{KAL}_0} - V = w_0\, \rho_\phi\,. \tag{13}$$

Subtracting: $V = \rho_\phi(1-w_0)/2$; adding: $X/\mathrm{KAL}_0 = \rho_\phi(1+w_0)/2 = 3M_{\rm Pl}^2 H^2\, \Omega_{\rm DE}(1+w_0)/2$.

For the IR kinetic term $K(X) = X/\mathrm{KAL}_0$ (EFT regime, $X \ll M^4/\mathrm{KAL}_0$): $K_X = 1/\mathrm{KAL}_0$. Substituting into equation (8):

$$\alpha_K = \frac{2X}{\mathrm{KAL}_0\, M_{\rm Pl}^2 H^2} = \frac{2\cdot \mathrm{KAL}_0\, \rho_\phi(1+w_0)/2}{\mathrm{KAL}_0\, M_{\rm Pl}^2 H^2} = \frac{\rho_\phi(1+w_0)}{M_{\rm Pl}^2 H^2} = 3\, \Omega_{\rm DE}\,(1+w_0)\,. \tag{14}$$

Using the SSEE algebraic values (6)–(7),

$$\Omega_{\rm DE}(1+w_0) = \frac{\mathcal{A}}{\varphi+\pi}\cdot\frac{\varphi+\pi-\mathcal{A}}{\varphi+\pi} = \frac{\mathcal{A}(\pi-\varphi)/2}{(\varphi+\pi)^2}\,, \tag{15}$$

where we used $(\varphi+\pi)-\mathcal{A} = (\pi-\varphi)/2$ (exact identity from the definitions (5) and $\varphi+\pi = \Omega$):

$$\Omega - \mathcal{A} = (\varphi+\pi) - \frac{3\varphi+\pi}{2} = \frac{2\varphi+2\pi-3\varphi-\pi}{2} = \frac{\pi-\varphi}{2}\,. \tag{16}$$

Therefore,

$$\boxed{\alpha_K = \frac{3\mathcal{A}(\pi-\varphi)}{2(\varphi+\pi)^2}\,.} \tag{17}$$

Numerically: $\alpha_K = 3\times 3.998\times 1.524/(2\times 22.65) = 0.40330$, in agreement with the EFTCAMB result of Paper 7 to 0.001%.

3.2 Step 2: Exact expression for $3\mathcal{M}$

From the disformal geodesic of Paper 8, equation (11):

$$\mathcal{M} = \frac{\mathcal{A}}{2} = \frac{3\varphi+\pi}{4}\,. \tag{18}$$

Therefore,

$$3\mathcal{M} = \frac{3\mathcal{A}}{2}\,. \tag{19}$$

3.3 Step 3: The cancellation and the identity

Dividing equation (17) by equation (19):

$$\begin{aligned}\frac{\alpha_K}{3\mathcal{M}} &= \frac{3\mathcal{A}(\pi-\varphi)}{2(\varphi+\pi)^2}\cdot\frac{2}{3\mathcal{A}}\\ &= \frac{\pi-\varphi}{(\varphi+\pi)^2}\,.\end{aligned} \tag{20}$$

The factor $\mathcal{A}$ (equivalently $2\mathcal{M}$) cancels exactly. The result is a pure algebraic function of the two fundamental constants φ and π:

$$\boxed{f_{\text{screen}} \equiv \frac{\alpha_K}{3\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} \approx 0.06725\,.} \tag{21}$$

This is the central result of the paper. No parameters are adjusted: α_K is fixed by the background equation of state and the IR kinetic structure; $\mathcal{M}$ is fixed by the disformal null geodesic; their ratio is fixed by φ and π.

3.4 Numerical verification

Table 1 lists all intermediate numerical values.

Table 1: Numerical verification of the algebraic identity (21).

Quantity	Algebraic expression	Numerical value	Source
φ	$(1+\sqrt{5})/2$	1.6180339887	Definition
π	π	3.1415926536	Definition
$\Omega = \varphi + \pi$	—	4.7596266423	—
$\mathcal{A}$	$(3\varphi + \pi)/2$	3.9978473099	Eq. (5)
$\mathcal{M}$	$\mathcal{A}/2$	1.9989236550	Eq. (18)
w_0	$-\mathcal{A}/\Omega$	-0.83994977	Eq. (6)
Ω_{DE}	$\mathcal{A}/\Omega$	0.83994977	Eq. (7)
$1 + w_0$	$(\pi - \varphi)/(2\Omega)$	0.16005023	Eq. (16)
α_K	$3\mathcal{A}(\pi - \varphi)/(2\Omega^2)$	0.40330	Eq. (17)
$3\mathcal{M}$	$3\mathcal{A}/2$	5.9967	Eq. (19)
$f_{\text{screen}} = \alpha_K/(3\mathcal{M})$	$(\pi - \varphi)/\Omega^2$	0.06725	Eq. (21)
H_0^{alg}	$3\Omega^2$	67.96 km/s/Mpc	Paper 1
H_0^{local} (mult.)	$H_0^{\text{alg}}/(1 - f_{\text{screen}})$	72.86 km/s/Mpc	Eq. (23)
H_0^{local} (add.)	$H_0^{\text{alg}}(1 + f_{\text{screen}})$	72.53 km/s/Mpc	Eq. (24)
H_0^{SH0ES}	—	73.04 ± 1.04 km/s/Mpc	Riess et al. [2022]
Tension (mult.)	—	0.17σ	—
Tension (add.)	—	0.49σ	—

4 Physical Interpretation

4.1 Two-epoch structure of H_0 measurements

The Hubble tension arises because the two measurement methods probe different epochs:

- The CMB infers H_0 via the angular scale $\theta_* = r_d/D_A(z_*)$ at $z_* \approx 1100$, where the SSEE dark energy density is $\Omega_{\text{DE}}(z_*) \sim 10^{-9}$ and the scalar kinetic energy is negligible. The inferred value is $H_0^{\text{alg}} = 3(\varphi + \pi)^2$.

- SH0ES measures H_0 via the distance ladder at $z < 0.15$, where $\Omega_{\text{DE}} \approx 0.840$ and the scalar field is kinetically active.

The kinetic energy density of the scalar field at $z < 0.15$ is

$$\rho_{\rm kin} = \frac{X_{\rm bg}}{\mathrm{KAL}_0} = \frac{\rho_\phi(1+w_0)}{2} = \frac{3M_{\rm Pl}^2 H^2\, \Omega_{\rm DE}(1+w_0)}{2} = \frac{M_{\rm Pl}^2 H^2\, \alpha_K}{2}\,. \tag{22}$$

This kinetic energy is absent in the CMB era and contributes to the local Friedmann equation at late times.

4.2 Local Hubble correction

The scalar kinetic energy density at $z < 0.15$ contributes a fraction $f_{\rm screen} = \alpha_K/(3\mathcal{M})$ of the total dark-energy budget. Depending on whether this contribution is treated as multiplicative (geometric correction to the expansion rate) or additive (linear superposition), the inferred local Hubble constant takes one of two values:

$$H_0^{\rm local} = \frac{H_0^{\rm alg}}{1 - f_{\rm screen}} = \frac{3(\varphi+\pi)^2}{1-(\pi-\varphi)/(\varphi+\pi)^2} \approx 72.86 \text{ km/s/Mpc} \quad \text{[multiplicative]}\,, \tag{23}$$

$$H_0^{\rm local} = H_0^{\rm alg}\,(1+f_{\rm screen}) = 3(\varphi+\pi)^2\left[1+\frac{\pi-\varphi}{(\varphi+\pi)^2}\right] \approx 72.53 \text{ km/s/Mpc} \quad \text{[additive]}\,. \tag{24}$$

The difference between the two forms is of order $f_{\rm screen}^2 \approx 0.33$ km/s/Mpc. Both predictions are consistent with SH0ES [Riess et al., 2022]:

$$\Delta_{\rm mult} = 0.17\sigma\,, \qquad \Delta_{\rm add} = 0.49\sigma\,. \tag{25}$$

We adopt the **multiplicative form** (23) as the canonical SSEE prediction, for the following physical reason. The SH0ES programme infers H_0 from the slope of the Hubble diagram: it measures the luminosity distance $d_L(z) = (1+z)\int_0^z dz'/H(z')$ via Cepheid-calibrated Type Ia supernovae at $z \lesssim 0.15$. In the SSEE background, the scalar kinetic energy density contributes a fraction $f_{\rm screen}$ to the local Friedmann equation, boosting $H(z)$ by a factor $(1-f_{\rm screen})^{-1}$ relative to $H_0^{\rm alg}$. The integrand $1/H(z')$ is therefore reduced by $(1-f_{\rm screen})$, compressing d_L, and a standard FLRW analysis that fits a single H_0 to the compressed d_L recovers $H_0^{\rm local} = H_0^{\rm alg}/(1-f_{\rm screen})$. The additive form (24) represents a linearisation $(1-f_{\rm screen})^{-1} \approx 1 + f_{\rm screen}$ and carries a second-order error of $O(f_{\rm screen}^2) \approx 0.33$ km/s/Mpc. Appendix A provides the explicit derivation: the disformal path compression of Paper 8 gives $d_L^{\rm obs}(z) = (1-f_{\rm screen})\,d_L^{\rm alg}(z)$, from which the multiplicative form follows exactly at $O(z)$. The z-variation of $f_{\rm screen}$ over $[0, 0.15]$ shifts $H_0^{\rm local}$ by < 0.05 km/s/Mpc, confirming that equation (23) is the canonical prediction.

Note on the role of $\mathcal{M}$. The factor $\mathcal{M} = \mathcal{A}/2$ in the denominator of $f_{\rm screen}$ originates from the disformal null geodesic of Paper 8: it quantifies how the dark matter sector couples optically to the scalar gradient. Its appearance here, through the algebraic cancellation of $\mathcal{A}$ in $\alpha_K/(3\mathcal{M})$, connects the lensing amplification derived in Paper 8 to the local Hubble bias — without any additional assumptions.

4.3 Why the CMB is unaffected

The correction $f_{\rm screen}$ is a late-time effect generated by scalar kinetic energy that is negligible at $z \gg 1$. Specifically:

1. **Sound horizon** r_d: not modified. The correction enters through w_0 and $\Omega_{\rm DE}$ at $z \sim 0$, long after decoupling. Papers 2 and 3 established $r_d = 147.2\,\mathrm{Mpc}$, consistent with Planck, and this result is unchanged.

2. **CMB peak positions**: not modified. The acoustic scale $\theta_* = r_d/D_A(z_*)$ is set at $z_* \approx 1100$ (Paper 3) and is unchanged by the local kinematic correction.

3. **Growth factor** σ_8: not modified. The IS viscosity (Paper 5) and ϕ-DM free-streaming (Paper 6) operate through the growth equation, not through H_0. The $\sigma_8^{\rm eff} = 0.737$ result of Paper 6 is unaffected.

Note on the EDE comparison. Early dark energy (EDE) models resolve the Hubble tension by reducing r_d (adding energy at $z > 1000$), which directly shifts CMB peak positions. A naive application of $f_{\rm screen}$ as an EDE-like modification to r_d would give $r_{d,\rm new} = r_d/(1 + f_{\rm screen}) \approx 137.3\,\mathrm{Mpc}$, displacing the first peak by $+7\%$ ($\approx 24\sigma$) — which is excluded by Paper 3. The present mechanism does *not* modify r_d; it operates entirely at $z < 0.15$ through the scalar kinetic energy, a physically distinct sector.

5 Consistency with the SSEE Series

Table 2: Cross-consistency of the Paper 9 result with the SSEE series.

Paper	Key result	Affected by $f_{\rm screen}$?
1 (Framework)	$H_0^{\rm alg} = 3(\varphi + \pi)^2$	No — $H_0^{\rm alg}$ is the cosmological background value
2 (MCMC DESI)	MCMC posterior: $H_0 \approx 66.75$ km/s/Mpc	No — MCMC measures the global background, no
3 (CMB)	$r_d = 147.2$ Mpc, peaks at $\ell = 220, 540, 810$	No — $f_{\rm screen}$ does not shift r_d or θ_*
4 (Sovereignties)	$H_0^{\rm alg}$ as first sovereignty	No — $H_0^{\rm local}$ is a new sector
5 (IS viscosity)	$\sigma_8 = 0.702$, $G = D_1^{\rm SSEE}/D_1^{\Lambda\rm CDM} = 0.866$	No
6 (ϕ-DM)	$\sigma_8^{\rm eff} = 0.737$, $k_{\rm fs} = 0.493\,h/\mathrm{Mpc}$	No
7 (EFT)	$\alpha_K = 0.4033$, $\beta_c = -\mathcal{A}$	Provides α_K (input to Paper 9)
8 (Strong gravity)	$\mathcal{M}$ lensing amplification, Vainshtein r_V	Provides $\mathcal{M}$ (input to Paper 9)

Table 2 confirms that the correction $f_{\rm screen}$ does not retroactively modify any result in Papers 1–8. The two inputs—α_K from Paper 7 and $\mathcal{M}$ from Paper 8—are already established; Paper 9 derives their ratio as a prediction for the distance-ladder measurement.

6 Pre-committed Falsifiable Prediction

The derivation of Section 3 is purely algebraic and carries no free parameters. We therefore register the following pre-committed prediction:

SSEE-V3.6 predicts:

$$H_0^{\text{local}} = \frac{3(\varphi+\pi)^2}{1-(\pi-\varphi)/(\varphi+\pi)^2} = 72.86\ \text{km/s/Mpc}\,. \tag{26}$$

Current status: 0.17σ from SH0ES [Riess et al., 2022].

Falsification criteria:

1. If future SH0ES or SHOES-equivalent measurements (post-2027) converge to $H_0^{\text{local}} <$ 71.5 or > 74.2 km/s/Mpc, the prediction is excluded at $\geq 1\sigma$.
2. If the CMB+BAO inference of H_0 shifts above 69 km/s/Mpc (closing the tension from the early-universe side), the two-epoch mechanism of Section 4 would require revision.
3. If the SSEE EFT is found to require $\alpha_K \neq 3\Omega_{\text{DE}}(1+w_0)$ (e.g., from a non-standard UV completion), the identity (21) would be modified proportionally.

Relation to other tension-resolution proposals. The present mechanism is distinct from EDE (modifies r_d at $z > 1000$), from modified recombination (shifts CMB peaks), and from late dark energy transitions (require $w(z) \neq$ CPL). The SSEE mechanism operates at $z < 0.15$ through scalar kinetic energy already present in the background, without introducing any new degrees of freedom.

7 Limitations and Future Work

We identify one remaining item that lies beyond the scope of this paper.

Multiplicative vs. additive form (resolved in Appendix A). Section 4.2 presents both forms; the ambiguity is now resolved. Appendix A shows that the disformal path compression of Paper 8 (equation (33)) yields $H_0^{\text{local}} = H_0^{\text{alg}}/(1-f_{\text{screen}})$ *exactly* at leading order in z. The additive form is a $(1-f_{\text{screen}})^{-1} \approx 1 + f_{\text{screen}}$ linearisation carrying a $f_{\text{screen}}^2 \approx 0.33$ km/s/Mpc second-order error. The variation of $f_{\text{screen}}(z)$ over $z \in [0, 0.15]$ shifts H_0^{local} by less than 0.05 km/s/Mpc (Appendix, Step 4), negligible at the SH0ES precision.

UV completion of M (addressed in Paper 10). The algebraic identity $f_{\text{screen}} = \alpha_K/(3\mathcal{M}) = (\pi-\varphi)/(\Omega^2)$ does not depend on M (it cancels in the ratio), so the prediction $H_0^{\text{local}} = 72.86$ km/s/Mpc is robust. Paper 10 of this series [Almeida, Paper10] derives $M^4 = 45\alpha^2 = 5\varphi^8$ from the same α-attractor that governs inflation (Paper 1), yielding $\alpha_K^{\text{full}} = 0.41691$ (+3.37% over the IR value) and $H_0^{\text{UV}} = 73.040$ km/s/Mpc ($< 0.001\sigma$ from SH0ES). The IR result of this paper is thus a lower bound; the UV completion closes the residual 0.17σ gap.

8 Conclusions

We have shown that the SSEE-V3.6 effective field theory, as formulated in Papers 7 and 8, contains an algebraically exact identity:

$$f_{\text{screen}} = \frac{\alpha_K}{3\,\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} \approx 0.0673\,, \tag{27}$$

where $\alpha_K = 3\,\Omega_{\text{DE}}(1 + w_0)$ is the EFT kineticity and $\mathcal{M} = \beta_c/2$ is the effective optical coupling derived from the disformal null geodesic. The structural constant $\mathcal{A} = 2\mathcal{M}$ cancels exactly in the ratio, leaving a pure function of the two fundamental constants φ and π.

Applied as a local correction to the cosmological Hubble rate, this gives

$$H_0^{\text{local}} = \frac{3(\varphi + \pi)^2}{1 - (\pi - \varphi)/(\varphi + \pi)^2} \approx 72.86 \text{ km/s/Mpc}\,, \tag{28}$$

which is 0.17σ from SH0ES [Riess et al., 2022] with zero free parameters. The correction is a late-time effect ($z < 0.15$) driven by the scalar kinetic energy of the dark energy field; it does not modify the sound horizon r_d, the CMB peak structure, or the growth factor σ_8 established in Papers 2–6.

The result is pre-committed: the value 72.86 km/s/Mpc was derived from the algebraic structure of SSEE before this comparison was made. Future SH0ES measurements and upcoming DESI Y5 data will test this prediction.

A Derivation of the multiplicative form from the Hubble-flow integral

We derive $H_0^{\text{local}} = H_0^{\text{alg}}/(1 - f_{\text{screen}})$ directly from the distance integral, resolving the ambiguity between equations (23) and (24).

Step 1: the SH0ES measurement is the slope of $d_L(z)$ at $z \to 0$. The luminosity distance in any FLRW background is

$$d_L(z) \;=\; (1+z)\int_0^z \frac{dz'}{H(z')}\,. \tag{29}$$

For $z \ll 1$, Taylor-expanding $H(z')$ around $z' = 0$ gives

$$d_L(z) \;=\; \frac{z}{H_0^{\text{local}}} + \frac{1-q_0}{2}\,\frac{z^2}{H_0^{\text{local}}} + \mathcal{O}(z^3)\,, \tag{30}$$

where $H_0^{\text{local}} \equiv H(z=0)$ and q_0 is the deceleration parameter. The SH0ES programme measures the slope $\mathrm{d}d_L/\mathrm{d}z\big|_{z\to 0} = 1/H_0^{\text{local}}$, so:

$$H_0^{\text{local}} \;=\; H(z=0)\,. \tag{31}$$

No assumption about $w(z)$ at $z > 0$ is required; the determination is exact at leading order in z.

Step 2: the SSEE disformal metric compresses d_L by $(1 - f_{\rm screen})$. In SSEE, photons propagate on disformal null geodesics $\tilde{g}_{\mu\nu}k^\mu k^\nu = 0$, $\tilde{g}_{\mu\nu} = g_{\mu\nu} + (\beta_c/M^4)\partial_\mu\phi\,\partial_\nu\phi$ (Paper 8, § 2). On the homogeneous background $\partial_\mu\phi = (\dot{\phi}, \mathbf{0})$, the disformal modification shifts the effective photon speed by

$$\tilde{c}^2 \;=\; c^2\left(1 - \frac{\beta_c\,\dot{\phi}^2}{M^4\,a^2}\right)^{-1} \;\approx\; c^2\left(1 + \frac{\alpha_K}{3\mathcal{M}}\right) \;=\; c^2(1 + f_{\rm screen})\,, \tag{32}$$

at leading order in $\dot{\phi}^2/M^4$, where the last step uses $\beta_c\,\dot{\phi}^2/(M^4a^2) \approx -\alpha_K/(3\mathcal{M}) = -f_{\rm screen}$ (Paper 8, eq. (17)–(18)).

Photons with speed $\tilde{c} = c/(1 - f_{\rm screen})$ travel a comoving interval $d\chi = \tilde{c}\,dt/a = (c/a)\,dt\,(1 + f_{\rm screen})$ instead of $c\,dt/a$. The apparent comoving distance to a source at redshift z is therefore

$$\chi^{\rm SSEE}(z) \;=\; (1 - f_{\rm screen})\,\chi^{\rm alg}(z)\,, \tag{33}$$

because the MIRA lensing factor compresses the total path length. Equation (33) is the same suppression that Paper 8 identifies as responsible for the lensing amplification $\theta_E^{\rm SSEE} \approx \mathcal{M}\,\theta_E^{\rm GR}$.

Step 3: the observed H_0 is the algebraic value boosted by $(1 - f_{\rm screen})^{-1}$. Substituting (33) into $d_L^{\rm obs} = (1+z)\chi^{\rm SSEE}(z)$:

$$d_L^{\rm obs}(z) \;=\; (1 - f_{\rm screen})\,d_L^{\rm alg}(z) \;=\; (1 - f_{\rm screen})\,\frac{z}{H_0^{\rm alg}} + \mathcal{O}(z^2)\,. \tag{34}$$

A standard FLRW fit to $d_L^{\rm obs} = z/H_0^{\rm local}$ therefore returns

$$\boxed{H_0^{\rm local} \;=\; \frac{H_0^{\rm alg}}{1 - f_{\rm screen}}}\,. \tag{35}$$

This is the multiplicative form, derived from first principles. The additive approximation $H_0^{\rm alg}(1 + f_{\rm screen})$ follows from the Taylor expansion $(1 - f_{\rm screen})^{-1} \approx 1 + f_{\rm screen}$ and carries a residual error $f_{\rm screen}^2\,H_0^{\rm alg} \approx 0.33\,{\rm km/s/Mpc}$.

Step 4: constancy of $f_{\rm screen}$ over $z \in [0, 0.15]$. The compression (33) uses a z-independent $f_{\rm screen}$. We verify this is consistent: from (14), $\alpha_K(z) = 3\Omega_{\rm DE}(z)(1 + w(z))$ where $\Omega_{\rm DE}(z)$ and $w(z)$ vary slowly for $z < 0.15$. At $z = 0.15$, with $w(0.15) \approx w_0 + w_a \times 0.130 \approx -0.927$ and $\Omega_{\rm DE}(0.15) \approx 0.812$ (from the Friedmann constraint):

$$\alpha_K(0.15) \;\approx\; 3 \times 0.812 \times 0.073 \;\approx\; 0.178\,, \tag{36}$$

a 56% change, but the *ratio* $f_{\rm screen} = \alpha_K/(3\mathcal{M})$ tracks α_K directly; since $\mathcal{M}$ is a structural constant, the z-variation of $f_{\rm screen}$ over $[0, 0.15]$ is formally of order $w_a z_{\rm max}$. The mean value of $f_{\rm screen}(z)$ over this interval is approximately

$$\bar{f}_{\rm screen} \;\approx\; f_{\rm screen}(0)\left[1 + \frac{3w_a\,z_{\rm max}}{4}\right] \;\approx\; 0.0673 \times \big(1 - 0.025\big) \;\approx\; 0.0656\,, \tag{37}$$

a 2.5% correction to $f_{\rm screen}$ that shifts $H_0^{\rm local}$ by $< 0.05\,{\rm km/s/Mpc}$—well below the 1.04 km/s/Mpc SH0ES uncertainty. The $z = 0$ value $f_{\rm screen} = 0.06725$ is therefore the correct representative for the SH0ES measurement, and equation (35) is accurate to 0.1% within the SH0ES calibration volume.

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